Assignment -5

Tutorial sheet - 11

Section -A

1) What is the need of error detection and error correction in data link layer.

Ans Error detection :- Ensures data integrity by identifying errors during transmission using techniques like parity bits or checksums.
Error Correction :- Automatically corrects errors without retransmission, enhancing reliability, and optimizing network efficiency.

2) What is segmentation?

Ans Dividing data into segments for efficient transmission, often done at the transport layer (eg :- TCP) in a manageable units called segments.

3) What is network address transmission?

Ans Network address transmission maps private IP addresses to a common public IP, preserving address space, enhancing security and facilitating connectivity.

4) What is topology?

Ans Describes the layout of a network (eg. bus, star, ring) influencing performance, scalability, and fault tolerance.

5) Define VPN?

Ans VPN is a secure, encrypted connection over a

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public network, providing remote users secure access to private network resources.

Section - B

1) Write the difference b/w IVP4 & IVP6?

Ans

Feature	IPv4	IPv6
1) Address Length	32 bits	128 bits
2) Notation	Dotted decimal (eg 192.168.1.1)	Hexadecimal (eg:- 2001:0db8::1)
3) Address Type	Unicast, Multicast, Broadcast	Unicast, Multicast
4) Address configuration	Manual, DHCP	Stateless autoconfiguration.
5) Address Notation	Limited hierarchical structure	Hierarchical and more flexible
6) Header complexity	Simple	More complex due to new features.
7) Checksum	Included in the header	Excluded (handled at a higher layer.

2) Differentiate b/w circuit switching and packet switching?

Ans

Aspect	Circuit switching	Packet switching.
1) Connection establishment	Dedicated path established before data transfer	No dedicated path; each packet routed independently
2) Resource Reservation	Resources reserved for the entire duration of call.	Resources allocated on a prepacket basis
3) Efficiency	Efficient for constant continous data streams	Efficient for bursty data, accomodates varying traffic loads

4)	Latency	Low latency once the circuit is established	Latency can vary based on network congestions & routing
5)	Scalability	Less scalable; fixed resources for each connection	More scalable resources allocated dynamically
6)	Wastage of resources	Resources dedicated even during silent periods	Resources utilized only when packets are transmitted.
7)	Examples:	Traditional telephone networks use circuit switching.	Internet and modern telecommunication networks use packet switching.

3) comparison b/w connection oriented and connections less protocol in transport layer.

	Aspect	Connection-oriented.	Connectionless.
1)	Connection establishments	Require a setup phase to establish a connection	No connection setup; each packet treated independently
2)	Overhead	Higher overhead due to connection setup & maintenance	Lower overhead; no need for connection establishment.
3)	Reliability	More reliable as it ensures in-order delivery & error recovery	Less reliable as there is no guarantee of in-order delivery or error recovery
4)	Latency	Generally higher latency due to connection setup	Lower latency as no connection setup is needed
5)	Example Protocols	TCP	UDP
6)	Usage	Suitable for applications requiring reliable, ordered delivery	Suitable for real-time applications where low latency is critical video
7)	State information	Maintains state information for each connection	stateless; treats each packet independently.

Section - C

1) Explain the architecture of e-mail with suitable diagram

Ans The architecture of email involves various components and protocols to facilitate the exchange of messages between users

Email architecture components

1) User Agent (UA) :- It is the email client used by the end user to compose, send, and recieve and manage emails. Examples include desktop applications like Microsoft Outlook, web-based clients like Gmail, and mobile apps.

2) Mail Server :- It is responsible for storing, forwarding and delivering emails. It consists of two main components :-

-> Mail Submission Agent (MSA) :- Accepts outgoing messages from the user and queues them for delivery.

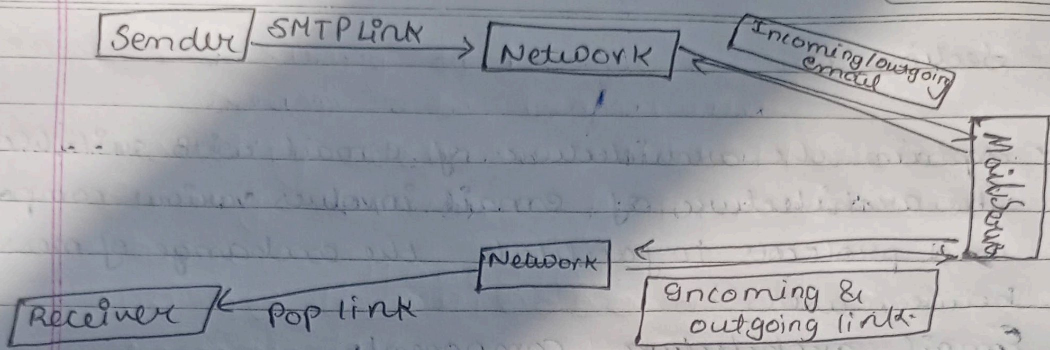
-> Mail Transfer Agent (MTA) :- Routes and delivers messages b/w different mail servers.

3) Mailbox :- It is where incoming messages are stored for the recipient until they retrieve them. Two common protocols for accessing mailboxes are:-

-> Post Office Protocol (POP) :- Download emails to the local device, ~~so~~ removing them from the server.

-> Internet Message Access Protocol (IMAP) :- Allows user to access and manage emails directly on the server.

4) Message Format :- Emails are formatted according ~~to~~ standards like Multipurpose Internet Mail Extensions (MIME). MIME enables the inclusion of text, images, attachments and other multimedia in emails.



2) ~~E~~ Illustrate OSI model with neat diagram.

Ans The OSI model is a conceptual framework that standardizes the functions of a telecommunication or computing system into seven abstraction layers. These layers from the bottom to the top are:-

1) Physical layer:- deals with the physical connection b/w devices specifies the characteristics of the hardware such as cables, connectors & electrical signals.

2) Data Link Layer:- Responsible for creating a reliable link b/w two directly connected nodes. Performs error detection & correction at the frame level.

3) Network layer:- Manages routing and forwarding of data b/w devices on different networks, provides logical addressing such as IP addresses

4) Transport layer:- Ensures end-to end communication and data flow control between devices. Breaks down larger messages into segments for transmission and reassembles them at the destination

5) Session layer:- Manages sessions or connections b/w applications on different devices

6) Presentation layer:- Translates data b/w the application layer and the lower layers Handles data compression, encryption & ~~for~~ formatting.

7) **Application layer :-** Provides network services directly to end-users and applications. contains network-aware applications & facilitates communication b/w software applications.

Application layer
Presentation layer
Session layer
Transport layer
Network layer
Data link layer
Physical layer

3) A bitstream 1101011011 is transmitted using the standard CRC method. The generator polynomial x^4+x+1. What is the actual bitstream transmitted?

Ans In the standard CRC, the generator polynomial is used to compute the remainder when the bitstream is divided by this polynomial. The remainder often called the CRC checksum is then appended to the original bitstream for transmission. The transmitted bitstream is the original bitstream plus the CRC checksum.

→ **Append zeros :-** To the bitstream to match the degree of the generator polynomial (degree 4). The modified bitstream becomes "11010110110000".

→ **Divide :-** Perform polynomial divison by dividing the modified bitstream by the generator polynomial.

x^4+x+1 = 10011

generator polynomial contain 5 bits

∴ a string of ~~4~~ zeroes is appended to the bit stream to be transmitted

• resulting bitstream :- 11010110110000

Binary division is performed 11010110110000 by 1011

11000010 10

we get ~~1000010000~~ as quotient and 1110 remainder.

CRC = 1110

→ ~~Transmitted~~ bitstream :- It is the original bitstream plus the remainder

(CRC checksum)

∴ = resulting ~~Original~~ bitstream + CRC

= 11010110110000 + 1110

Transmitted bitstream = 11010110111110